

QUESTION 1: Working with Lists

1. Create a list called numbers containing at least **8 integers**.
2. Using a **for loop**, do the following:
 - Print all numbers in the list.
 - Print the sum of all numbers.
 - Print only the even numbers.

Requirements:

- Use at least one loop.
- Use conditional statements.
- Display clear output.

QUESTION 2: Dictionary Basics

1. Create a dictionary called student with the following keys:
 - name
 - age
 - course
 - marks (must store a list of 3 marks)
2. Using loops:
 - Print all keys and their values.
 - Calculate the average of the marks.
 - Display whether the student **Passed** (average ≥ 50) or **Failed**.

Requirements:

- Dictionary must contain a list.
- Use loops to calculate the average.

QUESTION 3: List of Dictionaries

1. Create a list called employees.
2. Each employee must be stored as a dictionary containing:
 - name
 - department
 - salary
3. Using loops:
 - Print details of all employees.
 - Find the employee with the highest salary.
 - Calculate the total salary expense.

Requirements:

- Minimum 3 employees.
- Use comparison logic.
- Use loops to process data.

QUESTION 4: While Loop with User Input

1. Ask the user to enter numbers.
2. Store the numbers in a list.
3. Stop when the user enters -1.
4. After the loop ends:
 - Print the complete list.
 - Print the largest number.
 - Print the smallest number.

Requirements:

- Must use a **while loop**.
- Must store input in a list.
- Handle user input correctly.

QUESTION 5: Word Frequency Counter

1. Ask the user to enter a sentence.
2. Convert the sentence into a list of words.
3. Use a dictionary to count how many times each word appears.
4. Print the word frequency clearly.

Example Format (for understanding only):

apple: 3
banana: 2
orange: 1

Requirements:

- Use string splitting.
- Use a dictionary for counting.
- Use loops for processing.

QUESTION 6: Nested Loops – Multiplication Table

Using **nested loops**, print a multiplication table from 1 to 5.

Expected Output Format:

```
1 2 3 4 5
2 4 6 8 10
3 6 9 12 15
4 8 12 16 20
5 10 15 20 25
```

Requirements:

- Must use nested loops.
- Proper formatting required.

General Requirements

- Use:
 - At least 2 lists

- At least 2 dictionaries
 - At least 3 loops
 - At least 1 nested loop
- Include comments explaining your code.
- Proper indentation is required.
- Code must run without errors.
- Submit your .py file before the due date.